4.1 Interpreting the Derivative in Context

Calculus

Practice

For each problem, a differentiable function is given along with a definition of the variables. Interpret the values in the context of the problem.

1. The percentage grade a student receives on a test, is modeled by $G(t)$ where t is the number of hours spent studying for the test. Interpret $G'(1) = 3$.

   At 1 hour of studying, the grade will improve at a rate of 3% per hour.

2. Mr. Bean rides his motor scooter to work some days. His distance from home can be modeled by $d(t)$ meters where t is measured in minutes. Interpret $d'(15) = 650$.

   At 15 minutes, the distance from home is increasing at a rate of 650 meters per minute.

3. The rate at which a factory produces baseball hats each day can be modeled by $b(t)$ where $b(t)$ is the number hats produced per hour and t is the number of hours since the factory opens. Interpret $p'(1) = 100$.

   At 1 hour, the rate of producing hats is increasing at a rate of 100 hats per hour per hour.

4. Mr. Brust has entered a Biggest Loser contest and is hoping to lose some of those holiday calories. His weight gain or loss can be modeled by $p(t)$, where p is measured in pounds per week and t is weeks since he started his diet. Interpret $p'(4) = -1$.

   On the 4th week, the rate of weight loss is decreasing at a rate of 1 pound per week per week.

5. The number of gallons of water in a storage tank at time t, in minutes, is modeled by $w(t)$. Interpret $w'(10) = -8$.

   At 10 minutes, the amount of water is decreasing at a rate of 8 gallons per minute.

6. The rate at which the temperature is changing on a given day is modeled by $T(h)$, where T is measured in degrees per hour and h is hours. Interpret $T'(20) = -0.5$.

   At 20 hours, the rate of temperature change is decreasing at a rate of 0.5 degrees per hour per hour.

7. A harbor's water depth changes with the ocean tides. The rate of change of the depth of the water is modeled by $d(t)$, where d is measured in feet per hour and t is hours. Interpret $d'(2) = -3$.

   At 2 hours, the rate of change of the water depth is decreasing at a rate of 3 feet per hour per hour.

8. The height of a rocket is modeled by $h(t)$ meters where t is measured in seconds. Interpret $h'(10) = 30$.

   At 10 seconds, the height is increasing at a rate of 30 meters per second.

9. The time it takes for a chemical reaction to occur can be modeled by $t(A)$, where t is the time, in minutes, and A is the catalyst used, measured in milliliters. Interpret $t'(40) = 1.7$.

   When 40 milliliters of catalyst is used, the reaction is increasing at a rate of 1.7 minutes per milliliter.

10. A tire is leaking air pressure because of a small hole. The function $p(t)$ models the amount of air pressure (psi) in the tire after t minutes. Interpret $p'(3) = -2$.

    At 3 minutes, the air pressure is decreasing at a rate of 2 psi per minute.

4.2 Position, Velocity, and Acceleration

Calculus

Solutions **Practice**

1. A particle moves along a line so that its position at any time $t \geq 0$ is given by the function

$$s(t) = \frac{1}{3}t^3 - 3t^2 + 8t - 5$$

where s is measured in meters and t is measured in seconds.

a. Find the instantaneous velocity at any time t.

$v(t) = t^2 - 6t + 8$

b. When is the particle at rest?

$v(t) = 0$

$(t-4)(t-2) = 0$

$t = 2$ and $t = 4$ seconds

c. Find the acceleration of the particle at any time t.

$a(t) = 2t - 6$

d. What is the displacement of the particle for the first 3 seconds?

$s(0) = -5$

$s(3) = 9 - 27 + 24 - 5 = 1$

$s(3) - s(0) =$ 6 meters

2. A ball is dropped off a 1200 ft cliff. The height of the ball over time is modeled by the function $h(t) = 1200 - 16t^2$ where h is the height of the ball and t is time in seconds.

a. Find $h'(3)$. Explain what it means.

$h'(t) = -32t$

$h'(3) = -96$

At 3 seconds, the ball is falling at a rate of 96 feet / second.

b. Find $h''(3)$. Explain what it means.

$h''(t) = -32$

$h''(3) = -32$

At 3 seconds, the rate the ball is falling is increasing by 32 feet / second per second.

3. The position, in meters, of a body at time t sec is $s(t) = t^3 - 6t^2 + 9t$. Find the body's acceleration each time the velocity is zero.

$v(t) = 3t^2 - 12t + 9$

$0 = 3(t^2 - 4t + 3)$

$0 = (t-3)(t-1)$

$t = 1 \quad t = 3$

$a(t) = 6t - 12$

$a(1) = 6 - 12 \qquad a(3) = 18 - 12$

$a(1) = -6$ m/sec^2 $\quad a(3) = 6$ m/sec^2

4. A particle P moves on the number line. The graph $s = f(t)$ shows the position of P as a function of time t.

a. When is P moving to the left? → $s'(t) < 0$

(2, 3) and (5, 6)

b. When is P moving to the right? → $s'(t) > 0$

(0, 1)

c. When is P at rest? → $s'(t) = 0$

(1, 2) and (3, 5)

d. Graph the particle's velocity where defined.



5. The figure shows the velocity $v = \frac{ds}{dt} = f(t)$ of a body moving along a coordinate line in meters per second.

a. When does the body reverse direction?

$t=2,\ t\approx 7.8$ Velocity changes sign

b. When is the body moving at a constant speed?

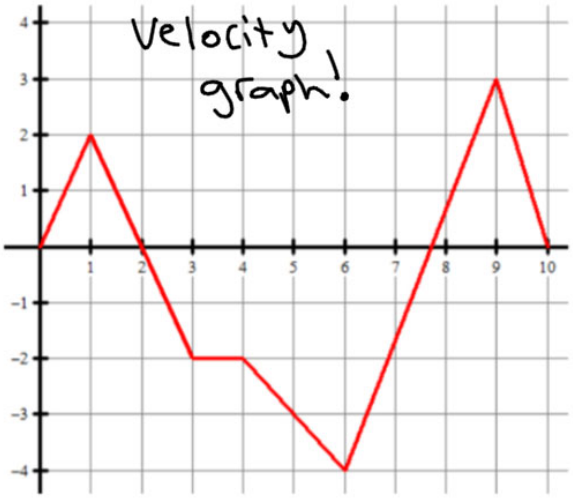
$(3,4)$

c. What is the body's maximum speed?

4 meters per second

d. What time interval(s) is the body speeding up?

$(0,1)\ (2,3)\ (4,6)\ (7.8,9)$



6. A rock thrown vertically upward from the surface of the moon at a velocity of 32 meters per second reaches a height of $s(t) = 32t - 0.8t^2$ meters in t seconds.

a. Find the rock's velocity and acceleration as functions of time.

$v(t) = 32 - 1.6t$

$a(t) = -1.6$

b. How long did it take the rock to reach its highest point?

at rest means $v(t)=0$.

$32 - 1.6t = 0$

$t = 20$ seconds

7. The data in the table gives selected values for the velocity, in meters per minute, of a particle moving along the x-axis. The velocity v is a differentiable function of time t.

Time t	0	2	5	6	8	12
Velocity $v(t)$	-3	2	3	5	7	5

a. At $t = 0$, is the particle moving to the right or left? Justify.

Left because $v(0)<0$.

b. Is there a time during the time interval $0 \le t \le 12$ minutes when the particle is at rest? Justify.

Yes, between 0 and 2 minutes. By the Intermediate Value Theorem (IVT), v(t) must equal zero in that interval.

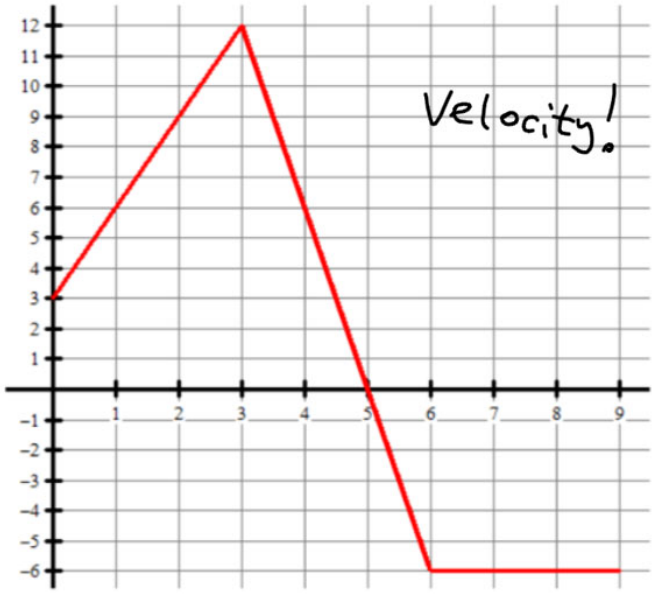
c. Use the data from the table to approximate $v'(10)$.

$\frac{v(12)-v(8)}{12-8} = \frac{-2}{4} = -\frac{1}{2}$

d. Explain the meaning of $v'(10)$ in terms of the particle motion.

At 10 minutes, the particle's velocity is decreasing by 0.5 meters per minute^2.

8. The graph represents the velocity, in feet per second, of a particle moving along the x-axis over the time interval from $t = 0$ to $t = 9$ seconds.



a. At $t = 4$, is the particle moving to the right or left? Justify.

Right because $v(4) > 0$.

b. Over what time interval is the particle moving left? Justify.

$(5, 9)$ because $v(t) < 0$.

c. At $t = 4$, is the acceleration positive or negative? Justify.

Negative. $v'(4) < 0$.

d. What is the average acceleration of the particle over the interval $2 \le t \le 4$? Show the computations and label your answer.

$$\frac{v(4) - v(2)}{4 - 2} = \frac{6 - 9}{2} = -\frac{3}{2} \text{ ft/sec}^2$$

e. At what time t in the given interval is the particle furthest to the right. Justify.

t=5. The particle travels right from (0, 5) seconds, then travels left for (5, 9) seconds.

---

9. A particle moves along the x-axis so that at time t its position is given by

$$x(t) = t^3 - 6t^2 + 9t + 11$$

where t is measured in seconds and x is measured in feet.

a. At $t = 0$, is the particle moving to the right or left? Justify.

$x'(t) = 3t^2 - 12t + 9$

$x'(0) = 9$

Right because $x'(0) > 0$

b. At $t = 1$, is the velocity of the particle increasing or decreasing? Justify.

$x''(t) = 6t - 12$

$x''(1) = 6 - 12 = -6$

Decreasing because $x''(1) < 0$.

c. What is the displacement over the first 6 seconds?

$x(0) = 11$

$x(6) = 65$

$x(6) - x(0) =$ 54 feet

No test prep. These questions are already like exam type problems.

4.3 Rates of Change Other Than Motion

Calculus

Solutions **Practice**

1. A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function E defined by $E(t) = 0.3t^4 - 14t^3 + 110t^2$ for $0 \le t \le 12$. At what rate are shoppers entering the store 5 hours after the start of the sale?

$E'(5) = 200$ shoppers per hour

---

2. The function $t = f(P)$ models the time, in days, for a small pond to evaporate as a function of the size P of the pond, measured in liters. What are the units for $f''(P)$?

days/liter^2

3.

t (days)	$W(t)$ (°F)
0	58
5	60
10	61
15	64
20	68

The temperature, in degrees Fahrenheit (°F), of a lake is a differentiable function W of time t. The table above shows the water temperature as recorded every 5 days over a 20-day period. Use the data from the table to find an approximation for $W'(10)$. Show the computations that lead to your answer. Indicate units of measure.

$\frac{W(15) - W(5)}{15 - 5} = \frac{4}{10} = 0.4$ °F per day

---

4. The function $D(t) = 13 - 2.7\cos\left(\frac{\pi}{4}t\right)$ models the depth, in feet, of water t hours after 6 A.M. Find the instantaneous rate of change of the depth of the water at 9 A.M. Use appropriate units.

$D'(3) \approx 1.499$ ft/hour

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5. The rate of consumption of liquid chocolate in Switzerland is given by $R(t) = ce^{kt}$, where R is measured in millions of gallons per year and t is measured in years from the beginning of 1990. The consumption rate doubles every 9 years and the consumption rate at the beginning of 1990 was 2 million gallons per year. Find c and k.

$2 = ce^{0.077(0)}$

$2 = c$

Double → $2c = ce^{k(9)}$

$2 = e^{9k}$

$\ln(2) = 9k$

$k \approx 0.077$

6.

t (minutes)	0	10	20	30
$W(t)$ (°F)	104	90	79	72

The temperature of water in a bathtub at time t is modeled by a strictly decreasing, twice-differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. The water is cooling for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t are given in the table above. Use the data in the table to estimate $W'(15)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

$W'(15) \approx \frac{W(20) - W(10)}{20 - 10} = \frac{-11}{10} = -1.1$

The water in the bathtub is cooling at an estimated rate of -1.1 degrees Fahrenheit per minute at 15 minutes.

---

7. The rate of decay, in grams per minute, of a radioactive substance is a differentiable, decreasing function R of time, t, in minutes. The table below shows the decay rate as recorded every 4 minutes over a 24-minute period.

t (minutes)	0	4	8	12	16	20	24
$R(t)$ (grams per minute)	320	221	130	82	39	22	11

a. Approximate $R'(12)$. Show the computations that lead to your answer. Indicate units of measure.

One of these! →

$\frac{W(16) - W(8)}{16 - 8} = \frac{39 - 130}{8} = -11.375$ grams/min^2

$\frac{W(16) - W(12)}{16 - 12} = -10.75$ g/min^2

$\frac{W(12) - W(8)}{12 - 8} = \frac{82 - 130}{4} = -12$ grams/min^2

b. A physicist proposes the function $G(t) = 320(0.882)^t$ as a model for the rate of decay of the radioactive substance, using the same units as $R(t)$. Find $G'(12)$. Using appropriate units, explain the meaning of your answer in terms of the decay rate of the substance.

-8.905 grams/min/min. At time $t = 12$ minutes, the decay rate is decreasing at the rate of 8.905 grams per minute per minute.

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8. For time $t \geq 0$ hours, let $r(t) = 50\left(1 - e^{-0.7t^2}\right)$ represent the speed, in miles per hour, at which a car travels along a straight road. Find $r'(2)$. Indicate units of measure.

8.513 miles per hour^2

4.3 Rates of Change Other Than Motion

Test Prep

9. People are entering a museum at a rate modeled by $f(t)$ people per hour and exiting the building at a rate modeled by $g(t)$ people per hour, where t is measured in hours. The functions f and g are nonnegative and differentiable for all times t. Which of the following inequalities indicates that the rate of change of the number of people in the building is decreasing at time t?

When is the rate decreasing?

(A) $g(t) < 0$ (B) $g'(t) < 0$ (C) $f(t) - g(t) < 0$ (D) $f'(t) - g'(t) < 0$

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10. **Calculator Active.** The rate of $R(t)$ of a certain item produced in a factory is given by

$$R(t) = 4000 + 48(t-3) - 4(t-3)^3$$

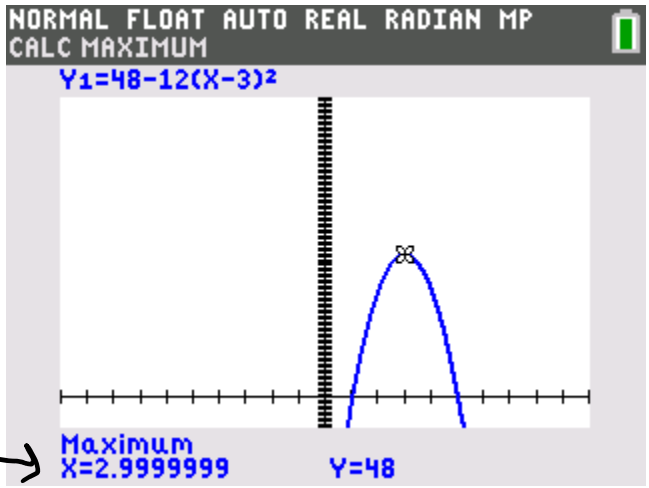
where t is the number of hours since the beginning of the workday at 8:00 a.m. At what time is the rate of production increasing most rapidly?

When is R(t) increasing most? = R'(t) has a max

$$R'(t) = 48 - 12(t-3)^2$$

(A) 8:00 am (B) 10:00 am (C) 11:00 am (D) 1:00 pm

Graph R'(t) and find max.



t = 3

8 AM + 3 = 11 A.M

4.4 Introduction to Related Rates

Calculus

Solutions **Practice**

Find a relationship between the given rates of change by doing the following.

1. **Set up a relationship using variables for the situation.**
2. **Differentiate with respect to time *t*.**

1. An ice cube is melting. Relate the volume's rate of change with the edges' rate of change.



$V = s^3$

$$\frac{dV}{dt} = 3s^2 \frac{ds}{dt}$$

2. The width and length of a rectangle are increasing. Relate these rates of change with the rate of change of the area of the rectangle.

$A = L \cdot W$

$$\frac{dA}{dt} = \frac{dL}{dt} W + L \frac{dW}{dt}$$

3. A spherical balloon is expanding. Relate the rate of change of surface area with the rate of change of the radius of the balloon. $A = 4\pi r^2$

$$\frac{dA}{dt} = 8\pi r \frac{dr}{dt}$$

4. The water level is dropping in a cylindrical tank because of a small leak in the tank. Relate the rates of change between the water level and the volume of the water. The volume is modeled by $V = \pi r^2 h$.

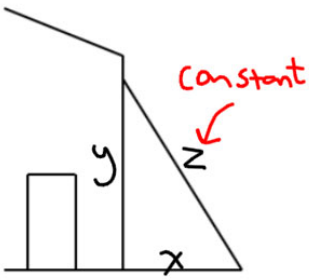
zero!

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt} h + \pi r^2 \frac{dh}{dt}$$

r is a constant

$$\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$$

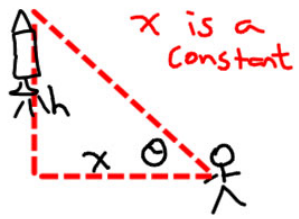
5. Mr. Brust is using a ladder to paint his house. The ladder is leaning against the house when Mr. Sullivan decides to pull the base of the ladder away from the house. Set up a relationship of the rates of change of how high the top of the ladder is and the distance from the bottom of the ladder to the house.



$x^2 + y^2 = z^2$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

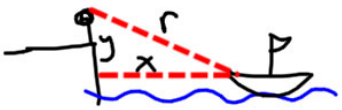
6. A rocket is rising vertically. An observer on the ground is standing a certain distance from the rocket's launch point. As the observer watches the rocket, the angle of elevation is increasing. Relate the rates of change of the angle of elevation with the speed of the rocket.



$\tan\theta = \frac{h}{x} = \frac{1}{x}h$

$\sec^2\theta \frac{d\theta}{dt} = \frac{1}{x}\frac{dh}{dt}$

7. A boat is being pulled toward a dock by a rope attached to its bow through a pulley on the dock. The pulley is higher than the boat's bow by several feet. Relate the rate that the rope is hauled in with how fast the boat is approaching the dock.



y is a Constant

$x^2 + y^2 = z^2$

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2z\frac{dz}{dt}$

zero

$2x\frac{dx}{dt} = 2z\frac{dz}{dt}$

4.4 Introduction to Related Rates

Test Prep

8. Let a and b be functions of time t such that the sum of a and three times b is constant. Which of the following equations describes the relationship between the rate of change of a with respect to time and the rate of change of b with respect to time?

$a + 3b = c$

$\frac{da}{dt} + 3\frac{db}{dt} = 0$

(A) $\frac{da}{dt} = 3\frac{db}{dt}$

(B) $\frac{da}{dt} = -3\frac{db}{dt}$

(C) $3\frac{da}{dt} + \frac{db}{dt} = 0$

(D) $\frac{da}{dt} + 3\frac{db}{dt} = F$, where F is a function of t.

---

9. The number of minutes M that it takes to make a calculus lesson and the number of lessons L that are made per week satisfy the relationship $L = \frac{k}{M}$, where k is a constant. Which of the following best describes the relationship between the rate of change, with respect to time t, of L and the rate of change, with respect to time t, of M?

$L = kM^{-1}$

$\frac{dL}{dt} = -kM^{-2}\frac{dM}{dt}$

(A) $\frac{dL}{dt} = \frac{k}{\left(\frac{dM}{dt}\right)}$

(B) $\frac{dL}{dt} = \frac{-k}{\left(\frac{dM}{dt}\right)}$

(C) $\frac{dL}{dt} = \frac{k}{M^2}\left(\frac{dM}{dt}\right)$

(D) $\frac{dL}{dt} = \frac{-k}{M^2}\left(\frac{dM}{dt}\right)$

4.5 Solving Related Rates

Calculus

Solutions **Practice**

1. If $y = 3x^4 + 6x$, find $\frac{dy}{dt}$ when $x = 1$, and $\frac{dx}{dt} = -3$.

$\frac{dy}{dt} = 12x^3\frac{dx}{dt} + 6\frac{dx}{dt}$

$\frac{dy}{dt} = 12(1)^3(-3) + 6(-3)$

$\frac{dy}{dt} = -36 - 18$

$\boxed{\frac{dy}{dt} = -54}$

2. If $g = 5h - h^5$, find $\frac{dg}{dt}$ when $h = 2$, and $\frac{dh}{dt} = 3$.

$\frac{dg}{dt} = 5\frac{dh}{dt} - 5h^4\frac{dh}{dt}$

$\frac{dg}{dt} = 5(3) - 5(2)^4(3)$

$\frac{dg}{dt} = 15 - 240$

$\boxed{\frac{dg}{dt} = -225}$

3. If $x^2 + y^2 = z^2$, find $\frac{dy}{dt}$ when $x = 3$, $y = 4$, $\frac{dx}{dt} = -1$, and $\frac{dz}{dt} = 5$.

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2z\frac{dz}{dt}$

$3^2 + 4^2 = z^2$

$25 = z^2$

$5 = z$

$2(3)(-1) + 2(4)\frac{dy}{dt} = 2(5)(5)$

$-6 + 8\frac{dy}{dt} = 50$

$\boxed{\frac{dy}{dt} = 7}$

4. If $A = \frac{1}{2}bh$, find $\frac{dA}{dt}$ when $b = 7$, $h = 6$, $\frac{db}{dt} = 2$, and $\frac{dh}{dt} = -3$.

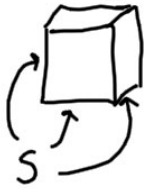
$\frac{dA}{dt} = \frac{1}{2}\frac{db}{dt}h + \frac{1}{2}b\frac{dh}{dt}$

$\frac{dA}{dt} = \frac{1}{2}(2)(6) + \frac{1}{2}(7)(-3)$

$= 6 - \frac{21}{2}$

$= \boxed{-\frac{9}{2}}$

5. An ice cube is melting at a rate of 5 cubic cm per hour. At what rate is the edge of the cube changing when the edge of the cube is 3 cm.



$\frac{dV}{dt} = -5$

$\frac{ds}{dt} = ?$

$s = 3$

$V = s^3$

$\frac{dV}{dt} = 3s^2\frac{ds}{dt}$

$-5 = 3(3)^2\frac{ds}{dt}$

$\boxed{\frac{ds}{dt} = -\frac{5}{27} \text{ cm/hour}}$

6. A circular pool of water is expanding at the rate of 16π in^2 / sec. At what rate is the radius expanding when the radius is 4 inches?



$\frac{dA}{dt} = 16\pi$

$\frac{dr}{dt} = ?$

$r = 4$

$A = \pi r^2$

$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$

$16\pi = 2\pi(4)\frac{dr}{dt}$

$\boxed{\frac{dr}{dt} = 2 \text{ in/sec}}$

7. A spherical balloon is expanding at a rate of 60π in^3 / sec. How fast is the surface area of the balloon expanding when the radius of the balloon is 4 inches? $V = \frac{4}{3}\pi r^3$ and $A = 4\pi r^2$.

$\frac{dV}{dt} = 60\pi$

$\frac{dA}{dt} = ?$

$r = 4$

$A = 4\pi r^2$

$\frac{dA}{dt} = 8\pi r \frac{dr}{dt}$?

$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

$60\pi = 4\pi (4)^2 \frac{dr}{dt}$

$\frac{15}{16} = \frac{dr}{dt}$

$\frac{dA}{dt} = 8\pi(4)\left(\frac{15}{16}\right)$

$\frac{dA}{dt} = 30\pi$ in^2/sec

8. An airplane (pt. A) is flying 600 mph on a horizontal path that will take it directly over an observer (pt. O). The airplane maintains a constant altitude of 7 miles (see figure). What is the rate of change of the distance between the observer and the airplane when $x = 5$ miles?

$\frac{dx}{dt} = 600$

$\frac{dy}{dt} = 0$

$\frac{dz}{dt} = ?$

$x = 5$

$5^2 + 7^2 = z^2$

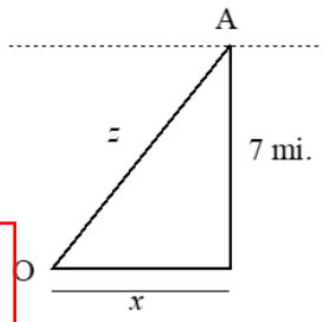
$\sqrt{74} = z$

$x^2 + y^2 = z^2$

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2z\frac{dz}{dt}$

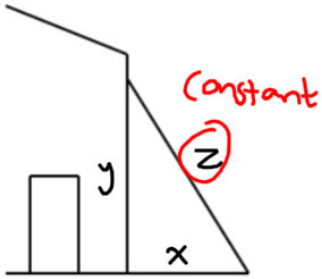
$2(5)(-600) + 0 = 2\sqrt{74}\frac{dz}{dt}$

$\frac{dz}{dt} = -\frac{3000}{\sqrt{74}}$ mph



constant!

9. Mr. Brust is using a ladder to paint his house. The 17-ft ladder is leaning against the house when Mr. Kelly decides to pull the base of the ladder away from the house at a rate of 3 ft./sec. How fast is the top of the ladder moving down the side of the house when it is 8 ft. above the ground? Indicate units of measure.



$z = 17$

$\frac{dx}{dt} = 3$

$\frac{dy}{dt} = ?$

$y = 8$

$x^2 + 8^2 = 17^2$

$x = 15$

$x^2 + y^2 = 17^2$

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

$2(15)(3) + 2(8)\frac{dy}{dt} = 0$

$\frac{dy}{dt} = -\frac{45}{8}$ ft/sec

10. A boat is being pulled toward a dock by a rope attached to its bow through a pulley on the dock. The pulley is 7 feet higher than the boat's bow. If the rope is hauled in at a rate of 4 feet/sec, how fast is the boat approaching the dock when 25 feet of rope is out?



$y = 7$

$\frac{dr}{dt} = -4$

$\frac{dx}{dt} = ?$

$r = 25$

$x^2 + 7^2 = 25^2$

$x = 24$

$x^2 + 7^2 = r^2$

$2x\frac{dx}{dt} + 0 = 2r\frac{dr}{dt}$

$2(24)\frac{dx}{dt} = 2(25)(-4)$

$\frac{dx}{dt} = -\frac{25}{6}$ ft/sec

4.5 Solving Related Rates

Test Prep

11. The base of a triangle is decreasing at a constant rate of 0.2 cm/sec and the height is increasing at 0.1 cm/sec. If the area is increasing, which answer best describes the constraints on the height h at the instant when the base is 3 centimeters?

$A = \frac{1}{2}bh$ $\quad \frac{dh}{dt} = 0.1$

$\frac{db}{dt} = -0.2$ $\quad \frac{dA}{dt} > 0$

$b = 3$

$\frac{dA}{dt} = \frac{1}{2}\frac{db}{dt}h + \frac{1}{2}b\frac{dh}{dt}$

$\frac{1}{2}(-0.2)h + \frac{1}{2}(3)(0.1) > 0$

$-0.1h > -0.15$

(a) $h > 3$ (b) $h < 1$ (c) $h > 1.5$ (d) $h < 1.5$ (e) $h > 2$

4.6 Approximating with Local Linearity

Calculus

Solutions

Practice

For each differential equation, let $y = f(x)$ be the particular solution to the differential equation with the given initial condition.

1. $\frac{dy}{dx} = (5 - y)\sin x$ and $f\left(\frac{\pi}{2}\right) = 2$.

   a. Write an equation for the line tangent to the graph of f at the point $\left(\frac{\pi}{2}, 2\right)$.

   $m = (5-2)\sin\left(\frac{\pi}{2}\right) = 3$

   $y - 2 = 3\left(x - \frac{\pi}{2}\right)$

   or

   $\boxed{y = 3x - \frac{3\pi}{2} + 2}$

   b. Use the tangent line to approximate $f(1.5)$.

   $\boxed{1.7876}$

2. $\frac{dy}{dx} = -\frac{4x}{y}$ and $f(1) = 3$.

   a. Write an equation for the line tangent to the graph of f at the point $(1,3)$.

   $m = -\frac{4}{3}$

   $y - 3 = -\frac{4}{3}(x - 1)$

   $\boxed{y = -\frac{4}{3}x + \frac{13}{3}}$

   b. Use the tangent line to approximate $f(1.1)$.

   $\boxed{2.8667}$

Answer the questions for each function listed.

3. $f(x) = 2\cos x + 1$ is concave down on $\left[0, \frac{\pi}{2}\right]$.

   a. What is the estimate for $f(1)$ using the local linear approximation for f at $x = \frac{\pi}{2}$? Give an exact answer (no rounding).

   $f\left(\frac{\pi}{2}\right) = 2\cos\left(\frac{\pi}{2}\right) + 1 = 1 \leftarrow y_1$

   $f'\left(\frac{\pi}{2}\right) = -2\sin\left(\frac{\pi}{2}\right) = -2 \leftarrow m$

   $y - 1 = -2\left(x - \frac{\pi}{2}\right)$

   $y - 1 = -2\left(1 - \frac{\pi}{2}\right)$

   $y - 1 = -2 + \pi$

   $\boxed{y = \pi - 1}$

   b. Is it an underestimate or overestimate? Explain.

   Overestimate because $f(x)$ is concave down

4. $f(x) = \frac{e^{2x}}{x+1}$ is concave up on $x > -1$.

   a. What is the estimate for $f(0.1)$ using the local linear approximation for f at $x = 0$?

   $f(0) = \frac{e^0}{0+1} = 1 \leftarrow y_1$

   $f'(x) = \frac{2e^{2x}(x+1) - e^{2x}(1)}{(x+1)^2}$

   $f'(0) = \frac{2e^0(1) - e^0}{(1)^2} = 1 \leftarrow m$

   $y - 1 = 1(x - 0)$

   $y = x + 1$

   $\boxed{y = 1.1}$

   b. Is it an underestimate or overestimate? Explain.

   Underestimate because $f(x)$ is concave up.

5. $f(x) = -\sqrt{4-x}$ is concave up on its domain.

a. What is the estimate for $f(1.9)$ using the local linear approximation for f at $x = 2$? Round to three decimal places.

$f(2) = -\sqrt{2}$ ← y_1

$f'(x) = -\frac{1}{2\sqrt{4-x}}(-1) = \frac{1}{2\sqrt{4-x}}$

$f'(2) = \frac{1}{2\sqrt{2}}$ ← m

$y + \sqrt{2} = \frac{1}{2\sqrt{2}}(x-2)$

$y = -1.4495$

b. Is it an underestimate or overestimate? Explain.

Underestimate. $f(x)$ is concave up.

6. f is concave down and $f(3) = -1$ and $f'(3) = 2$.

a. What is the estimate for $f(3.2)$ using the local linear approximation for f at $x = 3$?

$y+1 = 2(x-3)$

$y = 2x - 7$

$y = 2(3.2) - 7$

$y = -0.6$

b. Is it an underestimate or overestimate? Explain.

Overestimate. $f(x)$ is concave down.

7. f is concave up and $f(-5) = 2$ and $f'(-5) = -1$.

a. What is the estimate for $f(-5.1)$ using the local linear approximation for f at $x = -5$?

$y - 2 = -1(x+5)$

$y = -x - 3$

$y = 5.1 - 3$

$y = 2.1$

b. Is it an underestimate or overestimate? Explain.

Underestimate. $f(x)$ is concave up.

8. f is concave down and $f(2) = 1$ and $f'(2) = -3$.

a. What is the estimate for $f(1.9)$ using the local linear approximation for f at $x = 2$?

$y - 1 = -3(x-2)$

$y = -3x + 7$

$y = -3(1.9) + 7$

$y = 1.3$

b. Is it an underestimate or overestimate? Explain.

Overestimate. $f(x)$ is concave down.

4.6 Approximating with Local Linearity

$f(1) = 3 - 4 + 2 = 1$

Test Prep

9. Let f be the function given by $f(x) = 3x^2 - 4x + 2$. The tangent line to the graph of f at $x = 1$ is used to approximate values of $f(x)$. Which of the following is the smallest value of x for which the error resulting from this tangent line approximation is more than 0.5?

[*Hint for your calculator use*: Create a table to compare values of two functions.]

$f'(x) = 6x - 4$

$f'(1) = 2$

$y - 1 = 2(x-1)$

$y = 2x - 1$

(A) 1.3 (B) 1.4 (C) 1.5 (D) 1.6 (E) 1.7

10. The depth of snow in a field is given by the twice-differentiable function S for $0 \le t \le 12$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S, at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \le t \le 12$.

t (hours)	0	1	4	9	12
$S'(t)$ (centimeters per hour)	1.8	2.4	2.0	1.6	1.3

a. Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.

$S''(10) \approx \frac{S'(12) - S'(9)}{12 - 9} = \frac{-0.3}{3} = -0.1 \text{ cm/hr}^2$

The rate of the depth of snow is decreasing by 0.1 cm per hour per hour.

b. Is there a time t, for $0 \le t \le 12$, at which the depth of snow is changing at a rate of 1.5 centimeters per hour? Justify your answer?

Yes, on the interval $9 \le t \le 12$. By the Intermediate Value Theorem the rate must be 1.5.

c. At time $t = 4$, the depth of snow is 28 centimeters. Use the line tangent to the graph of S at $t = 4$ to approximate the depth of the snow at time $t = 6$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 6$? Justify your answer.

$S - 28 = 2(t - 4)$

$S = 2t + 20$

$S = 32$ cm

32 cm is an overestimate because $S(t)$ is concave down

4.7 L'Hospital's Rule

Calculus

Solutions **Practice**

Find the following. Use L'Hôpital's when possible.

1. $\lim_{x\to 1}\frac{x-1}{x^2-3x+2} = \frac{0}{0}$

$\lim_{x\to 1}\frac{1}{2x-3}$

$\frac{1}{2-3} = \boxed{-1}$

2. $\lim_{x\to -5}\frac{x^2-2x-35}{x+5} = \frac{25+10-35}{0} = \frac{0}{0}$

$\lim_{x\to -5}\frac{2x-2}{1}$

$-10-2 = \boxed{-12}$

3. $\lim_{x\to 0}\frac{4x}{\ln(x+1)} = \frac{0}{0}$

$\lim_{x\to 0}\frac{4}{\frac{1}{x+1}}$

$\frac{4}{1} = \boxed{4}$

4. $\lim_{x\to 0}\frac{x-1}{x^2-3x+2}$

Direct substitution!

$\boxed{-\frac{1}{2}}$

5. $\lim_{x\to 1}\frac{2(x^2-1)}{\ln x^2} = \frac{0}{0}$

$\lim_{x\to 1}\frac{4x}{\frac{2}{x}}$

$\frac{4}{2} = \boxed{2}$

6. $\frac{d}{dx}\frac{6x^2+x}{\sin(x)}$ Quotient!

$\boxed{\frac{(12x+1)(\sin x)-(6x^2+x)\cos x}{\sin^2 x}}$

7. $\lim_{x\to 0}\frac{2x^2}{e^x-1-x} = \frac{0}{0}$

$\lim_{x\to 0}\frac{4x}{e^x-1} = \frac{0}{0}$

$\lim_{x\to 0}\frac{4}{e^x}$

$\frac{4}{1} = \boxed{4}$

8. $\lim_{x\to 0}\frac{2x^2}{1-\cos(4x)} = \frac{0}{0}$

$\lim_{x\to 0}\frac{4x}{4\sin(4x)} = \frac{0}{0}$

$\lim_{x\to 0}\frac{4}{16\cos(4x)} = \boxed{\frac{1}{4}}$

9. $\lim_{x\to 0}\frac{\sqrt{4+x}-2}{x} = \frac{0}{0}$

$\lim_{x\to 0}\frac{\frac{1}{2\sqrt{4+x}}}{1}$

$\frac{1}{2\sqrt{4}} = \boxed{\frac{1}{4}}$

10. $\lim_{x\to -3}\frac{x-1}{x^2+7x+10}$

Direct substitution

$\frac{-3-1}{9-21+10} = \frac{-4}{-2}$

$\boxed{2}$

11. $\lim_{x\to\infty}\frac{e^{2x}}{2x^2} = \frac{\infty}{\infty}$

$\lim_{x\to\infty}\frac{2e^{2x}}{4x} = \frac{\infty}{\infty}$

$\lim_{x\to\infty}\frac{4e^{2x}}{4} = \boxed{\infty}$

12. $\lim_{x\to 0}\frac{\cos x-1}{x^2} = \frac{0}{0}$

$\lim_{x\to 0}\frac{-\sin x}{2x} = \frac{0}{0}$

$\lim_{x\to 0}\frac{-\cos x}{2} = \boxed{-\frac{1}{2}}$

13. $\frac{d}{dx}\frac{6x^2+x}{x+1}$ Quotient!

$\frac{(12x+1)(x+1)-(6x^2+x)(1)}{(x+1)^2}$

$\boxed{\frac{6x^2+12x+1}{(x+1)^2}}$

14. $\lim_{x\to\infty}\frac{\ln x^2}{\ln(x+4)^3}$ Use prop. of Logs

$\lim_{x\to\infty}\frac{2\ln x}{3\ln(x+4)}$

$\boxed{\frac{2}{3}}$

15. $\lim_{x\to -2}\frac{x+2}{x^2+2x-3}$ Direct subst.

$\frac{-2+2}{4-4-3} = \frac{0}{-3}$

$\boxed{0}$

4.7 L'Hospital's Rule

Test Prep

16. If $f(x) = 2x^3 + 5$, then $\lim_{x\to 0} \frac{f(x)-f(0)}{x^3}$ is $= \frac{0}{0}$

$\lim_{x\to 0} \frac{f'(x)}{3x^2} = \lim_{x\to 0} \frac{6x^2}{3x^2} = 2$

(A) 0 (B) 1 (C) 2 (D) 3 (E) The limit does not exist.

---

17. Functions f, g, and h are twice-differentiable functions with $g(3) = h(3) = 5$. The line $y = 5 + \frac{1}{2}(x-3)$ is tangent to both the graph of g at $x = 3$ and the graph of h at $x = 3$.

a. Find $h'(3)$.

Slope of h at $x=3$ is $\frac{1}{2}$

b. Let a be the function given by $a(x) = 2x^3 h(x)$. Write an expression for $a'(x)$. Find $a'(3)$.

$a'(x) = 6x^2 h(x) + 2x^3 h'(x)$

$a'(3) = 6(9)h(3) + (54)h'(3)$

$54(5) + 54(\frac{1}{2}) = \boxed{297}$

c. The function h satisfies $h(x) = \frac{x^2-9}{1-(f(x))^3}$ for $x \neq 3$. It is known that $\lim_{x\to 3} h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim_{x\to 3} h(x) = 5$ to find $f(3)$ and $f'(3)$. Show the work that leads to your answers.

$1-(f(3))^3 = 0$

$[f(3)]^3 = 1$

$\boxed{f(3) = 1}$

$\lim_{x\to 3} \frac{2x}{-3[f(x)]^2 \cdot f'(x)} = 5$

$\frac{2(3)}{-3[1]^2 \cdot f'(3)} = 5$

$\frac{2}{-f'(3)} = 5$

$\boxed{-\frac{2}{5} = f'(3)}$